

**Financing New Ventures:
Startup Status, Lender Composition, and Loan Terms in the U.S.
SBA 7(a) Program**

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Data: U.S. Small Business Administration 7(a) FOIA loan-level records, FY2010-FY2025

Abstract

This study examines differences in approved loan amounts and initial interest rates between startup and established-business borrowers in the U.S. Small Business Administration (SBA) 7(a) program. Using loan-level administrative records for FY2010-FY2025, the analysis includes 681,992 approved loans, of which 138,054 are classified as startups and 543,938 as established businesses. Sequential regressions, lender fixed effects, and industry and fiscal-year heterogeneity tests are used to assess the sensitivity of startup-established financing differences to alternative conditional comparisons. In the common-sample loan-amount analysis, the baseline specification implies approximately 18.0% higher approved amounts for startups. The estimate changes to -7.1% after adjustment for processing method and to -13.7% after business type and collateral are also included. With lender fixed effects, the baseline difference falls to 4.8% and is not statistically significant, while the fully adjusted estimate indicates approximately 15.2% lower approved amounts for startups. Initial interest rates are approximately 0.205 percentage points lower for startups in the baseline specification without lender fixed effects and 0.104 percentage points lower in the fully adjusted specification. These differences do not persist within lenders: the fully adjusted lender fixed-effects estimate is +0.046 percentage points and is not statistically significant. The startup-loan amount relationship also varies significantly across industries and fiscal years. Because the data contain approved loans only and the analysis is observational, the estimates describe conditional financing-term differences rather than causal effects or differences in credit access. Overall, the findings show that startup-established financing differences within the SBA 7(a) program are highly sensitive to lender composition, program and loan characteristics, industry, and time.

Keywords: entrepreneurial finance; startup financing; SBA 7(a); small-business lending; loan pricing; government loan guarantees

JEL classifications: G21; G28; L26

1. Introduction

Financing is a central challenge in the creation and development of new firms. Young businesses generally have shorter operating histories, less established financial information, and fewer observable signals of repayment capacity than mature firms. These characteristics can increase information problems between entrepreneurs and external financiers and shape both the financing available to young firms and the terms on which financing is provided. Financial life-cycle theory therefore emphasizes firm age, size, and informational opacity as important dimensions of financing structure (Berger & Udell, 1998).

Firm age alone, however, is unlikely to explain observed financing terms. Asset structure, organizational form, collateral, lender relationships, lending technologies, and institutional arrangements can also shape entrepreneurial financing. Cassar (2004), for example, documents relationships between startup financing and characteristics such as firm size, asset structure, organizational form, and growth orientation. Berger and Udell (2006) further emphasize that small-business finance depends not only on borrower characteristics but also on the lending technologies and financial institutions through which credit is provided.

Government-guaranteed lending provides an important setting in which to examine these relationships. The U.S. Small Business Administration (SBA) 7(a) program operates through participating lenders under SBA program requirements and can finance purposes including working capital, equipment, real estate, refinancing, and changes of ownership (U.S. Small Business Administration, 2026a).

Consequently, observed 7(a) loan terms reflect the interaction of borrower characteristics, lender practices, program structure, and loan characteristics.

Existing research demonstrates the importance of SBA lending for small-business finance and economic outcomes. Craig, Jackson, and Thomson (2007) examine SBA-guaranteed lending and local economic growth. Brown and Earle (2017), using linked SBA, lender, and U.S. employer data, estimate effects of financial access on employment growth and report stronger estimated impacts for younger firms in their setting. Hackney (2023) studies government-guaranteed lending during the 2007-2009 financial crisis and documents relationships between SBA 7(a) lender presence and small-business lending. Research has also examined startup borrowers within SBA lending. Kang (2017), using 683 SBA 7(a) loan activities involving information-technology-based startups, studies soft information and lender characteristics in high-tech startup debt financing. Thus, the contribution of the present study is not the first use of SBA data to study startups, but a broad examination of financing terms across a large, multi-industry sample of approved startup and established-business 7(a) loans.

A central empirical issue is that an observed startup-established financing difference may depend strongly on the comparison being made. An unconditional difference can reflect differences in industry, geography, lenders, program processing methods, organizational form, collateral, and other loan characteristics. Moreover, some characteristics of the loan contract should not automatically be treated as conventional pre-existing confounders: they may themselves reflect lender screening, program structure, borrower financing needs, or the contracting process. For this reason, progressively adjusted specifications answer different conditional questions rather than successively revealing a single causal effect.

Research question. How do approved loan amounts and initial interest rates differ between startup and established-business borrowers within the SBA 7(a) program, and how sensitive are these differences to program and loan characteristics, lender composition, industry, and fiscal year?

The analysis uses SBA 7(a) administrative records covering FY2010-FY2025. The primary analytical sample contains 681,992 approved loans, including 138,054 loans classified as startups and 543,938 classified as established businesses. Because the data contain approved loans only, the study examines financing terms conditional on receiving an observed SBA 7(a) loan; it does not estimate differences in application success or access to credit.

This study contributes to the entrepreneurial-finance literature in three related ways. First, it documents how startup-established differences in approved SBA 7(a) loan terms change across sequential specifications rather than relying on a single fully adjusted comparison. Second, it introduces lender fixed effects to distinguish pooled differences from financing-term differences observed within lenders, showing that lender composition is particularly important for the interest-rate results. Third, it documents significant industry and temporal heterogeneity in the startup-loan amount relationship. Together, these analyses shift the focus from whether startups receive uniformly more or less favorable terms toward understanding how observed financing differences depend on the institutional and contractual context in which approved SBA loans are originated.

2. Literature Review and Conceptual Background

2.1 Financial life-cycle theory and startup finance

Berger and Udell (1998) develop a financial-growth-cycle perspective in which financing sources and structures vary with firm size, age, and informational characteristics. Younger and smaller firms tend to be more informationally opaque because they possess shorter histories and less publicly observable information. As firms develop, their financing possibilities may change alongside their information environment and organizational characteristics.

Berger and Udell (2006) broaden this perspective by emphasizing lending technologies and institutional structures. Their framework highlights that SME financing cannot be understood solely from borrower characteristics; the technologies lenders use to originate and monitor credit and the financial environment in which lenders operate also matter.

Cassar (2004) provides direct evidence concerning financing around business formation. The study examines how startup size, asset structure, organizational type, growth orientation, and owner characteristics relate to financing choices and financing magnitudes. The findings are consistent with the relevance of information asymmetry, agency considerations, and transaction costs in startup financing.

These perspectives imply that differences between startup and established borrowers may reflect more than business age itself. Loan size and pricing may also vary with industry, location, collateral, organizational form, lender practices, and program structure.

2.2 Government-guaranteed small-business lending

The SBA 7(a) program introduces another institutional layer into the financing relationship. SBA states that 7(a) is its primary business loan program and that borrowers apply through participating lenders rather than directly to SBA. Specific terms are negotiated between the borrower and participating lender subject to SBA requirements (U.S. Small Business Administration, 2026a).

This institutional structure matters for empirical interpretation. Financing outcomes observed in 7(a) data reflect both borrower characteristics and the structure of a government-guaranteed lending program. Consequently, patterns observed within the program should not automatically be generalized to conventional bank credit or entrepreneurial financing as a whole.

Prior studies show the relevance of SBA lending for broader firm outcomes. Brown and Earle (2017) analyze linked SBA, lender, and U.S. employer data and estimate positive employment effects associated with financial access in their research design. Hackney (2023) studies government-guaranteed lending during a financial crisis and reports that areas with greater SBA 7(a) lender presence experienced higher small-business loan volume and other economic differences during the 2007-2009 crisis.

Kang (2017) provides a particularly relevant entrepreneurial-finance connection. Using 683 SBA 7(a) loan activities involving information-technology-based startups, the study examines soft information and reports relationships between default outcomes, geographic proximity to lenders, and lender size. This supports the broader proposition that lender and information characteristics may matter when analyzing startup debt financing.

2.3 Firm age and interest-rate setting

A simple information-asymmetry argument could suggest that young firms should face higher borrowing costs. Recent research, however, indicates that observed pricing patterns need not be monotonic with firm age.

Cowling et al. (2026) explicitly examine whether financial life-cycle theory extends to loan interest-rate setting. Using 31,760 UK loan contracts issued under the Recovery Loan Scheme from April 2021 through May 2024, they find substantial variation across the firm size-age continuum and report that new firms within size classes tend to receive lower interest-rate offers than early-stage firms. Their setting and identification differ from the present study, but the finding is relevant to interpreting a negative startup-rate association without assuming a simple monotonic age-pricing relationship.

The current analysis does not test a specific lender-pricing mechanism. Instead, it asks whether startup and established SBA borrowers exhibit different observed initial rates and whether those differences remain after conditioning on measurable loan and program characteristics.

Financial life-cycle theory suggests that firm age and informational opacity can be associated with financing structure, while institutional and lending-technology perspectives imply that observed financing terms also depend on how credit is originated and structured. Accordingly, the empirical analysis does not impose a universal directional prediction for the startup coefficient. It evaluates whether financing terms differ between startup and established borrowers and whether those differences remain stable across alternative institutional and contractual comparisons.

3. Data and Sample

This study uses loan-level administrative data from the U.S. Small Business Administration 7(a) program. The analysis combines the SBA 7(a) FOIA datasets covering fiscal years 2010-2019 and 2020-present. The SBA open-data portal states that the 7(a) and 504 FOIA files are updated quarterly and provides a data dictionary for the files (U.S. Small Business Administration, 2026b). Because the latter file used for this study was available only through June 30, 2026, FY2026 is excluded to avoid treating an incomplete fiscal year as comparable with complete years. The primary study period therefore covers FY2010-FY2025.

The combined data contain information on gross approval amounts, initial interest rates, loan terms, borrower location, industry, business age, processing method, business type, collateral status, lender, and other loan characteristics. After restricting the data to complete fiscal years from 2010 through 2025, 893,265 records remained before applying the business-age classification.

The primary analysis distinguishes startups from established businesses using the SBA BusinessAge field. A startup is defined specifically as a borrower classified as “Startup, Loan Funds will Open Business.” This definition is used because the exact category appears in both source periods.

The established-business group consists of observations classified as “Existing or more than 2 years old,” “Existing, 5 or more years,” “Less than 3 years old but at least 2,” “Less than 4 years old but at least 3,” or “Less than 5 years old but at least 4.” Change-of-ownership observations, unanswered or missing business-age records, and other new-business categories whose definitions differ across reporting periods are excluded from the primary comparison.

Applying these criteria produces an analytical sample of 681,992 approved loans: 138,054 startup loans and 543,938 established-business loans. Startups therefore represent approximately 20.2% of the analytical sample.

4. Variables and Empirical Strategy

4.1 Outcomes and explanatory variable

The analysis considers two primary financing outcomes: gross approval amount and initial interest rate. Gross approval amounts are strongly right-skewed, so regression models use the natural logarithm of GrossApproval. Coefficients on startup status in log-loan models are converted to percentage differences using $100 \times [\exp(\beta) - 1]$.

Initial interest rates are analyzed in percentage-point units. Observations with missing or zero reported initial interest rates are excluded from interest-rate models. Zero reported loan terms are excluded when term information is required. The primary explanatory variable, StartupBinary, equals one for borrowers meeting the startup definition and zero for borrowers in the established-business comparison group.

4.2 Sequential specifications

The analysis distinguishes between two groups of covariates. Fiscal year, project jurisdiction, industry, and lender fixed effects are used to account for broad temporal, geographic, sectoral, and lender-specific differences. Processing method, business type, collateral status, loan amount, term, and fixed-versus-variable interest-rate status are treated separately as observed program or loan characteristics. These latter variables are not assumed to be exogenous confounders or pre-treatment characteristics. Some may reflect lender screening, borrower financing needs, program rules, or features determined jointly during the lending process. Accordingly, coefficient changes across specifications are interpreted as evidence of specification sensitivity and differences in conditional comparisons, not as mediation estimates or evidence that successive models recover a causal effect.

The loan-amount analysis begins with a baseline specification relating log gross approval amount to startup status while absorbing fiscal-year, project-jurisdiction, and two-digit NAICS industry fixed effects. Processing method, business type, and collateral status are then introduced sequentially using a common estimation sample. Standard errors are clustered by lender. Percentage differences are obtained as $100 \times [\exp(\beta) - 1]$.

The interest-rate analysis follows the same sequential approach. After the baseline fixed effects, the models progressively introduce processing method, business type, collateral status, log approved loan amount, loan term, and fixed-versus-variable interest-rate status. Coefficients are interpreted in percentage-point units.

4.3 Lender fixed effects

A second set of specifications adds lender fixed effects. These models compare startup and established-business loans while absorbing time-invariant lender-specific differences. Lender fixed effects change the nature of the comparison but do not eliminate borrower selection, omitted borrower characteristics, or other forms of endogeneity; the estimates therefore remain noncausal. Standard errors continue to be clustered by lender.

4.4 Industry and fiscal-year heterogeneity

Heterogeneity is evaluated by allowing the startup association in the loan-amount model to vary across two-digit NAICS industries and, separately, across fiscal years. Joint Wald tests assess whether the industry-specific and year-specific startup coefficients are equal. Individual industry estimates are treated descriptively, with the global tests providing the primary inference on heterogeneity.

4.5 Robustness and inflation adjustment

The analysis retains two robustness checks from the original specification: alternative clustering by project jurisdiction and a stricter established-business definition. It also adjusts annual loan amounts using Consumer Price Index for All Urban Consumers (CPI-U), U.S. city average, All Items annual averages from the U.S. Bureau of Labor Statistics (2026), and expresses descriptive loan amounts in 2025 dollars. Because the regression models include fiscal-year fixed effects, year-specific inflation adjustment is absorbed by those fixed effects on the log scale; its principal value is improving interpretation of long-run descriptive trends.

All regressions are observational. The startup coefficient is interpreted as a conditional association among approved SBA 7(a) loans, not as a causal effect of startup status or an estimate of access to credit.

5. Results

5.1 Descriptive evidence

Table 1. Descriptive Statistics by Firm Stage

Measure	Established	Startup
N	543,938	138,054
Mean gross approval	\$399,458	\$462,802
Median gross approval	\$150,000	\$200,000
Mean initial interest rate	7.30%	7.14%
Median initial interest rate	6.50%	6.25%
Mean term (months)	126	131
Median term (months)	120	120
Collateralized	73.6%	90.1%
Fixed rate	18.9%	20.0%
Corporation	88.5%	92.7%

Note. Sample consists of approved SBA 7(a) loans in FY2010-FY2025 retained under the primary startup-versus-established business-age definition.

The analytical sample contains 681,992 approved SBA 7(a) loans, including 138,054 startup loans and 543,938 established-business loans. Startup loans have a mean gross approval amount of \$462,802 compared with \$399,458 for established businesses, and medians of \$200,000 and \$150,000, respectively. Initial interest rates are slightly lower for startups in the raw data, while other financing characteristics differ more substantially; for example, 90.1% of startup observations are identified as collateralized compared with 73.6% of established-business observations.

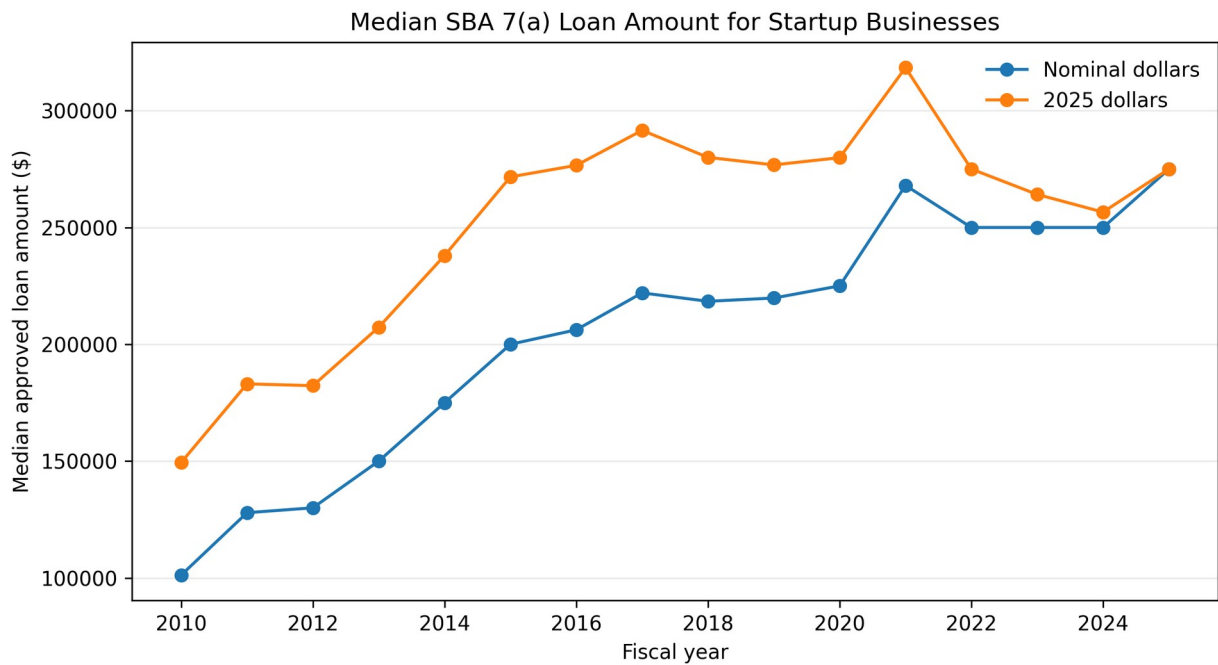
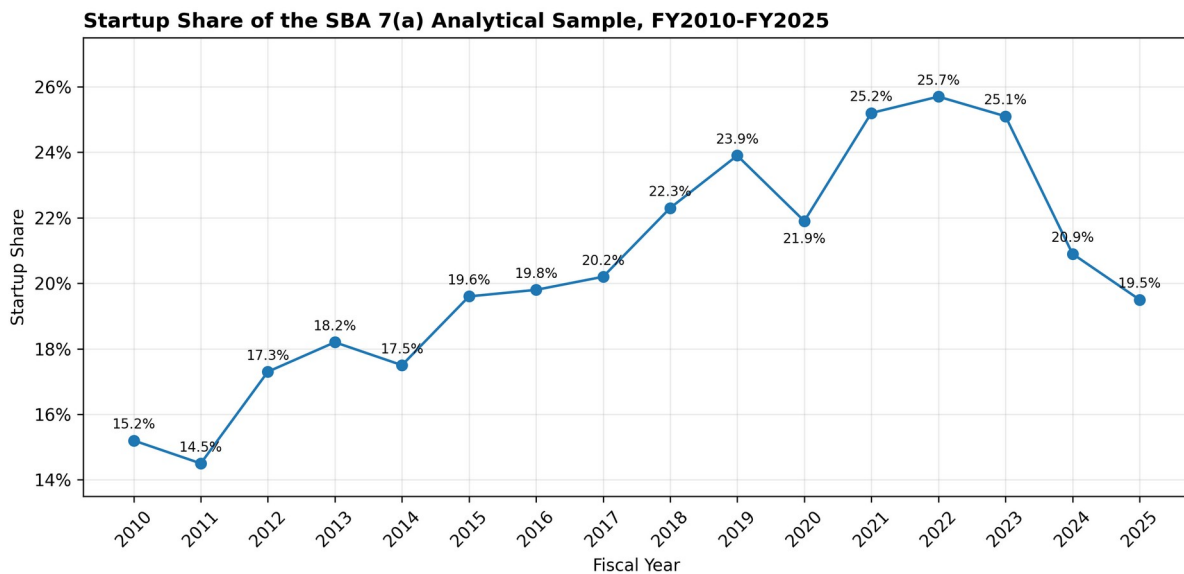


Figure 1. Median SBA 7(a) Loan Amount for Startup Businesses, FY2010-FY2025. Nominal values and CPI-U-adjusted values expressed in 2025 dollars. Source: author calculations using SBA 7(a) records and U.S. Bureau of Labor Statistics CPI-U annual averages.

Median startup approval increased from \$101,200 in FY2010 to \$275,000 in FY2025, a nominal increase of approximately 171.7%. Expressed in 2025 dollars, the FY2010 median is approximately \$149,414, implying a real increase of approximately 84.1% through FY2025. The real median reached approximately \$318,414 in FY2021 before declining and then returning to \$275,000 in FY2025.

Startup loans as a percentage of startup and established-business loans in the analytical sample



Source: U.S. Small Business Administration 7(a) FOIA data. Author's calculations.

Figure 2. Startup Share of the SBA 7(a) Analytical Sample, FY2010-FY2025. Startup share is calculated among observations classified as either startup or established businesses under the primary definition.

Startup representation also changes over time. The share was 15.2% in FY2010, reached 25.7% in FY2022, and was 19.5% in FY2025. These percentages describe the composition of the approved-loan analytical sample rather than startup prevalence among all businesses or applicants.

5.2 Sequential loan-amount specifications

Table 2. Sequential Loan-Amount Specifications

Specification	Startup beta	Clustered SE	p-value	Implied difference
Baseline	0.1657	0.0709	.019	+18.03%
+ Processing method	-0.0736	0.0606	.225	-7.10%
+ Business type	-0.0836	0.0599	.163	-8.02%
+ Collateral	-0.1478	0.0512	.004	-13.74%

Note. All specifications use the same analytical sample (N = 681,956) and include fiscal-year, project-jurisdiction, and two-digit NAICS industry fixed effects. Standard errors are clustered by lender. Percentage differences are calculated as $100 \times [\exp(\beta) - 1]$.

The baseline specification implies approximately 18.0% higher approved amounts for startups. Adding processing method changes the estimate to -7.1% and removes statistical significance. Adding business type produces comparatively little additional movement, while adding collateral yields an estimated -13.7% difference ($p = .004$). Because all models use the same observations, these coefficient movements are not attributable to changing sample composition. The largest initial shift occurs after adjustment for processing method, with an additional substantial shift after collateral.

5.3 Loan-amount estimates with lender fixed effects

Table 3. Loan-Amount Estimates with Lender Fixed Effects

Specification	Startup beta	Clustered SE	p-value	Implied difference
Baseline + lender FE	0.0468	0.0475	.325	+4.79%
+ Processing method	-0.1286	0.0302	<.001	-12.07%
+ Business type	-0.1338	0.0287	<.001	-12.52%
+ Collateral	-0.1654	0.0265	<.001	-15.24%

Note. Models absorb fiscal-year, project-jurisdiction, two-digit NAICS industry, and lender fixed effects. Standard errors are clustered by lender. N = 681,539 after fixed-effect singleton removal.

Once persistent lender-specific differences are absorbed, the baseline startup differential falls to approximately +4.8% and is not statistically significant ($p = .325$). After processing method is introduced, the estimate becomes -12.1%; business type changes the estimate only modestly, and the fully adjusted estimate is -15.2% ($p < .001$). The positive baseline differential is therefore not robust to lender fixed effects, whereas the fully adjusted negative loan-amount association remains in the within-lender specification.

5.4 Sequential initial interest-rate specifications

Table 4. Sequential Initial Interest-Rate Specifications

Specification	Startup coefficient (pp)	Clustered SE	p-value
Baseline	-0.2051	0.0818	.012
+ Processing method	-0.1229	0.0774	.112
+ Business type	-0.1181	0.0766	.123
+ Collateral	-0.0366	0.0550	.506
+ Loan amount	-0.1000	0.0422	.018
+ Term	-0.1051	0.0427	.014
+ Fixed/variable rate	-0.1044	0.0407	.010

Note. Coefficients are percentage-point differences in initial interest rates. All models include fiscal-year, project-jurisdiction, and two-digit NAICS industry fixed effects and cluster standard errors by lender. N = 681,703 after singleton removal.

The lower observed startup rate is sensitive to the conditioning set. Processing method attenuates the baseline coefficient from -0.205 to -0.123 percentage points and removes statistical significance. The coefficient is only -0.037 percentage points after collateral is added. Conditioning additionally on approved loan amount increases the negative magnitude to approximately -0.100 percentage points, after which term and fixed-versus-variable rate status produce relatively small changes.

5.5 Initial interest rates with lender fixed effects

Table 5. Initial Interest-Rate Estimates with Lender Fixed Effects

Specification	Startup coefficient (pp)	Clustered SE	p-value
Baseline + lender FE	-0.0188	0.0569	.741
+ Processing method	0.0756	0.0461	.101
+ Business type	0.0780	0.0456	.087
+ Collateral	0.1155	0.0390	.003
+ Loan amount	0.0510	0.0341	.135
+ Term	0.0473	0.0334	.157
+ Fixed/variable rate	0.0455	0.0327	.163

Note. Models include lender, fiscal-year, project-jurisdiction, and two-digit NAICS fixed effects. Standard errors are clustered by lender. N = 681,287.

The negative startup interest-rate association does not persist once lender fixed effects are introduced. The baseline within-lender coefficient is -0.019 percentage points ($p = .741$). The point estimate becomes positive in subsequent specifications, but after loan amount, term, and rate type are included, the fully adjusted estimate is +0.046 percentage points and is not statistically significant ($p = .163$). The evidence therefore does not support treating lower startup interest rates as a stable within-lender relationship.

5.6 Industry heterogeneity

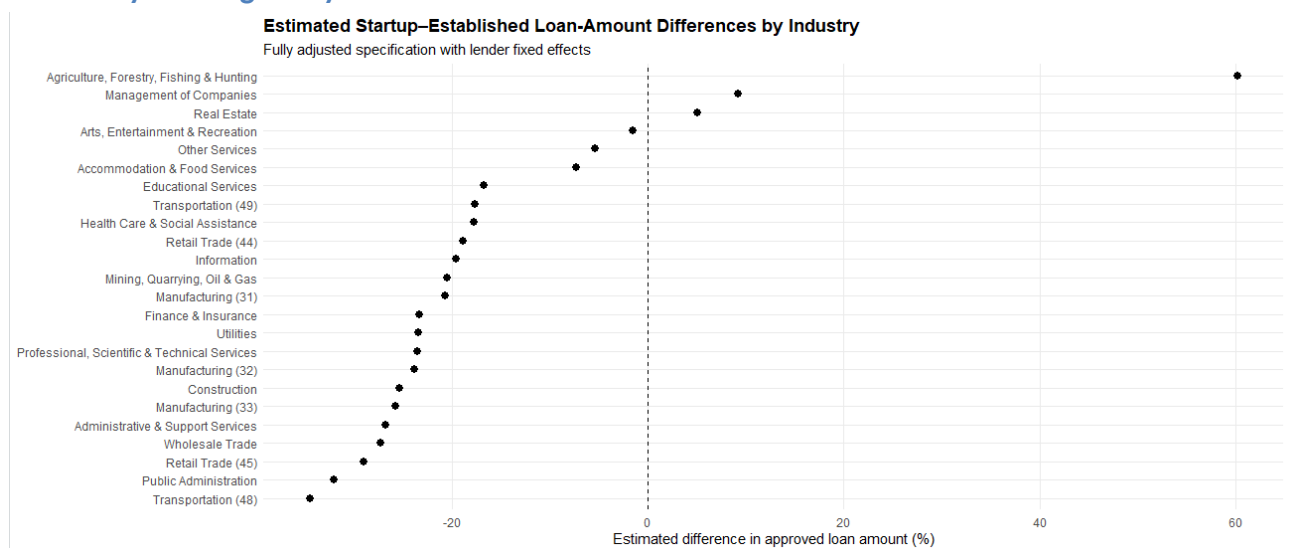


Figure 3. Estimated Startup–Established Loan-Amount Differences by Industry.

Note. Points show industry-specific startup coefficients from the fully adjusted loan-amount specification with fiscal-year, project-jurisdiction, industry, and lender fixed effects and controls for processing method, business type, and collateral status. Coefficients are transformed as $100 \times [\exp(\beta) - 1]$. The

joint test rejects equality of the startup coefficients across industries, Wald $F(23, 2611) = 16.00$, $p < .001$. Estimates are conditional associations; individual industry estimates are presented descriptively.

The startup-loan amount relationship varies substantially across industries. Industry-specific point estimates range from approximately 60.1% higher approved amounts in Agriculture, Forestry, Fishing and Hunting (NAICS 11) to approximately 34.4% lower amounts in Transportation (NAICS 48). Most point estimates are negative, while positive point estimates appear in NAICS 11, Real Estate (NAICS 53), and Management of Companies (NAICS 55). Because some industries contain relatively few startup observations, the joint test rather than separate industry-level significance tests provides the primary evidence of heterogeneity.

5.7 Fiscal-year heterogeneity

The startup-loan amount relationship also varies over time. A joint test of startup-by-fiscal-year interactions strongly rejects temporal constancy, Wald $F(15, 2611) = 8.34$, $p < .001$. The interaction coefficients are differences relative to FY2010 rather than standalone startup effects for each year, so the primary conclusion is that the conditional startup differential changes over FY2010-FY2025 rather than that any particular year has a separately identified startup premium or penalty.

5.8 Robustness checks

Table 6. Robustness of Extended-Model Startup Estimates

Specification	Log Gross Approval	Initial Interest Rate
Main model: lender clustering	-0.1478 (0.0512)	-0.1046 (0.0407)
Project-jurisdiction clustering	-0.1478 (0.0141)	-0.1046 (0.0217)
Strict business-age definition	-0.1779 (0.0521)	-0.0932 (0.0423)

Note. Standard errors in parentheses. The main and project-jurisdiction rows use the primary business-age definition; changing the clustering level changes estimated uncertainty, not coefficients. The strict business-age definition narrows the established comparison group.

Alternative clustering by project jurisdiction leaves the coefficient estimates unchanged but produces materially different standard errors, underscoring within-lender dependence. Under the stricter established-business definition, the extended loan-amount coefficient is -0.1779 (SE = 0.0521) and the rate coefficient is -0.0932 (SE = 0.0423), preserving the direction of the corresponding original extended-model estimates.

The inflation-adjusted fully adjusted lender fixed-effects loan-amount model produces $\beta = -0.16536$ (clustered SE = 0.02647, $p < .001$), corresponding to -15.24%, with $N = 681,539$. This is identical to the nominal specification because annual CPI adjustment is a fiscal-year-specific additive term on the log scale and is absorbed by the fiscal-year fixed effects.

6. Discussion

This study examines how approved loan amounts and initial interest rates differ between startup and established-business borrowers within the U.S. SBA 7(a) program. The expanded analysis shows that these differences cannot be summarized adequately by a single startup premium or penalty. Instead, the estimated relationships depend substantially on lender composition, program and loan characteristics, industry, fiscal year, and the particular conditional comparison being estimated.

The loan-amount results provide the clearest example. In the baseline specification with fiscal-year, jurisdiction, and industry fixed effects, startup status is associated with approximately 18.0% higher

approved loan amounts. This positive association reverses when additional characteristics are introduced sequentially. Adding processing method changes the estimate to -7.1%, adding business type produces comparatively little additional movement to -8.0%, and adding collateral status changes the estimate to -13.7%. The sequential analysis therefore shows that the reversal is concentrated primarily in the adjustment for processing method, with an additional substantial change following adjustment for collateral.

These coefficient movements should not be interpreted as a causal decomposition. Processing method and collateral are not necessarily conventional pre-existing confounders. They may reflect program structure, lender screening, borrower financing needs, or features of the lending process that are jointly related to observed loan terms. Consequently, the sequential models are best understood as alternative conditional comparisons that reveal how sensitive the startup coefficient is to the characteristics included in the comparison.

Lender composition is also important. When lender fixed effects are added to the baseline loan-amount specification, the estimated startup differential declines from 18.0% to 4.8% and is no longer statistically significant. Thus, the positive baseline relationship is not reproduced when persistent lender-specific differences are absorbed. In contrast, after processing method, business type, and collateral are included, the within-lender specification implies approximately 15.2% lower approved amounts for startup loans. The similarity between this estimate and the -13.7% estimate from the corresponding specification without lender fixed effects suggests that the negative fully adjusted loan-amount association is more robust than the positive baseline association.

The interest-rate results tell a different story. Without lender fixed effects, startup loans are associated with initial rates approximately 0.205 percentage points lower in the baseline model and 0.104 percentage points lower in the fully adjusted model. More importantly, the negative startup interest-rate association does not persist once lender fixed effects are introduced. The baseline within-lender coefficient is close to zero, and the fully adjusted coefficient is positive but statistically indistinguishable from zero. The results therefore caution against interpreting lower rates observed in pooled specifications as a general feature of startup borrowing within the 7(a) program.

The distinction between lender-clustered standard errors and lender fixed effects is important. Clustering allows for dependence among observations associated with the same lender and changes statistical inference, but it does not absorb systematic differences across lenders. Lender fixed effects instead absorb time-invariant lender-specific differences. The substantial coefficient changes after introducing lender fixed effects indicate that lender composition is an important dimension of observed startup-established financing differences.

The loan-amount relationship is also heterogeneous across the SBA 7(a) market. Joint tests reject equality of the startup relationship across industries and fiscal years. These results indicate that pooled national estimates mask meaningful sectoral and temporal variation and reinforce the need for caution when characterizing startup financing with a single average coefficient.

The descriptive time-series evidence provides an additional qualification. Median startup approval amounts increased substantially over FY2010-FY2025, but the increase is much smaller in real than nominal terms. Inflation adjustment does not alter the startup coefficient in models containing fiscal-year fixed effects, but it materially changes the interpretation of long-run descriptive growth.

Taken together, the findings are consistent with an institutional view of entrepreneurial lending in which observed financing terms reflect more than firm age alone. They do not establish why processing method,

collateral, lender composition, industry, or time are associated with coefficient changes. The data cannot distinguish borrower selection, lender screening, program design, financing needs, unobserved credit quality, or combinations of these mechanisms.

7. Limitations

First, the SBA administrative records used here contain approved loans, not the full population of firms seeking financing. The data do not include rejected applicants, firms seeking financing outside the SBA 7(a) program, or firms that chose not to apply. The analysis therefore concerns loan terms conditional on an observed 7(a) approval and cannot estimate whether startups are more or less likely than established firms to obtain credit.

Second, the analysis is observational. Fixed effects and measured covariates can account for particular observed or time-invariant differences, but they do not create random assignment of startup status. Startup and established borrowers may differ on dimensions unavailable in the SBA FOIA data, including credit scores, financial statements, profitability, owner wealth, lender-borrower relationships, and other indicators of credit quality.

Third, the interpretation of the extended specifications requires caution. Processing method, collateral, approved loan amount, term, and rate type can be features of the lending and contracting process rather than predetermined borrower characteristics. Their inclusion changes the conditional comparison but should not be interpreted mechanically as eliminating confounding. Coefficient changes across the sequential models are not formal mediation effects.

Fourth, lender fixed effects strengthen the comparison by absorbing persistent differences across lenders, but they do not solve all selection problems. The models rely on within-lender variation among approved loans and cannot account for time-varying lender practices or unobserved borrower characteristics. The lender identifier is based on BankName, as in the original analysis, rather than a separately validated persistent lender identifier.

Fifth, SBA BusinessAge categories changed across reporting periods. The primary analysis uses the exact startup category "Startup, Loan Funds will Open Business" and selected established-business categories while excluding categories that do not provide a sufficiently comparable definition. The stricter established-business sensitivity analysis helps assess this classification choice but does not eliminate potential measurement differences over time.

Sixth, the industry and fiscal-year analyses demonstrate statistically significant heterogeneity but do not establish the mechanisms producing that heterogeneity. Some industry cells are comparatively small, making individual sector estimates less precise. The joint heterogeneity tests are therefore more informative for the study's central conclusion than isolated significance tests for individual industries or years.

Seventh, inflation adjustment improves interpretation of loan amounts over the 16-year period but does not address changes in SBA program rules, lending environments, borrower composition, or macroeconomic conditions. Fiscal-year fixed effects absorb common year-specific differences in the regression models but do not identify which historical or policy changes produced the observed temporal heterogeneity.

Finally, the results are specific to SBA 7(a) lending and should not automatically be generalized to conventional bank lending, venture capital, private credit, or entrepreneurial finance outside government-guaranteed lending.

8. Conclusion

Using 681,992 approved SBA 7(a) loans from FY2010 through FY2025, this study examines differences in financing terms between startup and established-business borrowers and, more importantly, how those differences change across alternative empirical comparisons.

The expanded analysis shows that the positive baseline startup loan-amount differential is highly sensitive to specification. A baseline estimate of approximately 18.0% higher approved amounts reverses after adjustment for processing method and other loan characteristics. When lender fixed effects are introduced, the baseline difference declines to 4.8% and is not statistically significant, whereas the fully adjusted within-lender specification indicates approximately 15.2% lower approved amounts for startups.

The interest-rate evidence is less robust. Although specifications without lender fixed effects indicate modestly lower initial rates for startups, that relationship disappears in within-lender comparisons. The fully adjusted lender fixed-effects estimate is small and statistically indistinguishable from zero.

The study also finds substantial heterogeneity across industries and fiscal years, while inflation adjustment shows that the long-run increase in startup loan amounts is considerably smaller in real than nominal terms. Together, these results demonstrate that observed startup financing differences within the SBA 7(a) program depend importantly on who lends, how loans are processed and structured, the industry in which firms operate, and the period being examined.

Accordingly, the evidence does not support characterizing SBA startup financing with a universal startup premium or penalty. The more defensible conclusion is that startup-established differences among approved 7(a) loans are conditional and institutionally structured. Future research linking SBA administrative records to application outcomes, borrower financial characteristics, and more detailed lender information would be better positioned to distinguish selection, screening, program design, and borrower demand mechanisms.

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